Allometric, Endocrine, and Scanner Heterogeneity in Structural and Resting-State Neuroimaging of Gender Identity: A Hierarchical Bayesian Multivariate Meta-Analysis with Latent Covariate Imputation

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Abstract

Structural magnetic resonance imaging (sMRI), diffusion tensor imaging (DTI), and resting-state functional magnetic resonance imaging (rs-fMRI) investigations of gender identity have reported localized volumetric, microstructural, and connectivity differences relative to cisgender comparison groups matched by either birth-assigned sex or experienced gender identity. However, synthesis across this literature is complicated by three coupled methodological limitations: (i) inconsistent statistical adjustment for allometric brain scaling via total intracranial volume (TIV) and exogenous endocrine exposure via gender-affirming hormone therapy (GAHT), (ii) incomplete open reporting of subgroup-specific demographic covariates such as participant age, and (iii) widespread treatment of correlated multi-region effect sizes extracted from small single-scanner cohorts ($N = 10$–54 per group) as independent observations. Here we synthesize $k = 35$ standardized regional neuroimaging contrasts spanning $J = 18$ cortical, subcortical, and white-matter regions across $K = 11$ independent clinical imaging cohorts (12 primary publications, 2011–2020) using a joint hierarchical Bayesian multivariate meta-regression estimated via Hamiltonian Monte Carlo and the No-U-Turn Sampler (`CmdStan 2.39.0` / `brms 2.23.0`; 4 chains, 8,000 post-warmup draws, 0 divergent transitions, $\max \widehat{R} = 1.0019$). Rather than applying listwise deletion or single-value plug-in imputation where subgroup mean ages are absent from open records (4/35 contrasts, 11.4%), we calibrate a joint Gaussian measurement submodel on the 31 verified observed contrast ages ($\mu = 26.85$ years, SD $= 4.66$ years) and sample the unextracted covariates as latent parameters within the Markov chain. Under weakly informative regularizing priors ($b \sim \mathcal{N}(0, 0.5)$), all primary neuroanatomical contrasts yield 95% credible intervals spanning zero and intersecting the Region of Practical Equivalence (ROPE $= [-0.10, +0.10]$), including *Trans vs. Gender Identity* ($\beta = +0.1021$, 95% CrI $[-0.5095, +0.7065]$, 24.46% in ROPE, $p_{\text{Bayes}} = 0.7332$; covariate-adjusted marginal cell mean $d = +0.1227$, 95% CrI $[-0.5457, +0.8098]$) and *Trans vs. Natal Sex* ($\beta = -0.2706$, 95% CrI $[-0.8856, +0.3280]$, 17.20% in ROPE, $p_{\text{Bayes}} = 0.3802$; covariate-adjusted marginal cell mean $d = -0.2500$, 95% CrI $[-0.9423, +0.4425]$). Variance decomposition demonstrates that between-study/scanner heterogeneity ($\sigma_{\text{study}} = 1.0656$, 95% CrI $[0.6646, 1.7148]$) exceeds between-region anatomical heterogeneity ($\sigma_{\text{region}} = 0.0977$, 95% CrI $[0.0038, 0.3038]$, 61.11% inside ROPE) by an order of magnitude (posterior median ratio $\sigma_{\text{study}}/\sigma_{\text{region}} = 13.34$, 95% CrI $[3.02, 281.00]$; $\Pr(\sigma_{\text{study}} > \sigma_{\text{region}} \mid \mathcal{D}) = 1.0000$), accounting for 98.38% (95% CrI $[90.14\%, 100.00\%]$) of modeled hierarchical variance. Widening fixed-effect priors fourfold in variance to $\mathcal{N}(0, 1.0)$ confirms complete structural invariance ($\sigma_{\text{study}} = 1.0964$ vs. $\sigma_{\text{region}} = 0.0991$; median ratio 13.77). These findings indicate that reported regional neuroimaging differences in this corpus do not isolate

a localized, scanner-invariant neuroanatomical signature of gender identity once allometric, endocrine, demographic, and site-level variance components are jointly modeled.

1 Introduction

Over the past fifteen years, quantitative neuroimaging studies employing voxel-based morphometry (VBM), cortical volumetry, diffusion tensor imaging (DTI), structural connectomics, and resting-state functional connectivity (rs-fMRI) have sought to evaluate whether regional brain structure or intrinsic connectivity in transgender individuals systematically aligns with birth-assigned sex, shifts toward experienced gender identity, or exhibits distinct network organization [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]. Narrative reviews and mega-analytic syntheses have noted substantial heterogeneity across published findings, with individual single-site studies variously reporting shifts in white-matter fractional anisotropy or mean diffusivity [1, 2, 6, 10], subcortical or insular grey-matter volume differences [3, 9, 11], adolescent cerebellar and medial frontal differences [8], or altered fronto-parietal and salience-network coupling [5, 7, 12], alongside extensive cross-group overlap [19, 20, 21].

Evaluating these findings quantitatively requires confronting four statistical challenges well established in neuroimaging methodology:

1. **Small-Sample Winner's Curse and Type M/S Distortion:** As demonstrated across neuroscience broadly by Button et al. [17] and formalized via Type S (sign) and Type M (magnitude) error theory by Gelman and Carlin [23], single-site neuroimaging cohorts with group sizes of $N = 10$–30 have low statistical power to detect subtle population effects ($|d| \leq 0.30$). When cluster-forming or region-of-interest (ROI) selection thresholds are applied to noisy voxelwise maps, statistically significant survivors exhibit substantial magnitude inflation and sign instability.

2. **Allometric Head-Size Scaling (TIV) and Endocrine Exposure (GAHT):** Comprehensive meta-syntheses of human brain sex differences demonstrate that global brain size (total intracranial volume, TIV) accounts for the overwhelming majority of apparent regional volumetric dimorphism between cisgender men and cisgender women [18]. Simultaneously, cross-sex / gender-affirming hormone therapy (GAHT) directly modulates cortical thickness, ventricular volume, subcortical grey-matter volume, and white-matter water diffusivity [9, 11]. Studies that omit TIV normalization in diffusivity analyses or pool hormone-treated and hormone-naive cohorts risk confounding systemic allometry or exogenous steroid exposure with baseline neuroanatomy.

3. **Partial Covariate Missingness Across Primary Reports:** Cohort age spans from mid-adolescence (≈ 16 years [8]) to adulthood (25–35 years [3, 11]), yet open-access abstracts and data repositories occasionally omit subgroup-disaggregated mean ages for specific post-hoc contrasts. In conventional frequentist meta-regression, missing moderator values force either listwise deletion of entire studies—sacrificing statistical precision and inducing selection bias—or deterministic mean plug-in imputation, which understates standard errors by treating unobserved covariates as known constants.

4. **Crossed Hierarchical Dependence Between Scanners and Brain Regions:** Primary neuroimaging studies routinely report multiple contrasts (e.g., comparing a transgender cohort against both birth-assigned sex controls and gender-identity controls) across multiple

brain ROIs acquired on the same 3T scanner under identical pulse sequences and preprocessing pipelines. Ignoring within-study clustering conflates site, scanner, recruitment, and pipeline variance (σ_{study}) with reproducible neuroanatomical localization (σ_{region}).

To address all four challenges within a unified probabilistic framework, we conduct a hierarchical Bayesian multivariate meta-analysis across $k = 35$ extracted regional contrasts from $K = 11$ clinical cohorts ($J = 18$ brain regions) implemented in `Stan` [15] via `brms` [14]. Specifically, we (i) verify primary bibliographic provenance and backfill exact sample-size-weighted contrast ages from open full-text records where documented, (ii) jointly model the remaining unextracted ages as latent parameters via Bayesian measurement-error/missing-value imputation (`brms::mi()`), (iii) evaluate parameter posteriors against Kruschke's Region of Practical Equivalence (ROPE = $[-0.10, +0.10]$) [16], (iv) partition hierarchical variance between scanner/study cohorts and anatomical regions, and (v) verify robustness under a fourfold prior-variance sensitivity expansion.

2 Materials and Methods

2.1 Corpus Provenance, Contrast Coding, and Demographic Backfill

The empirical dataset (`trans_empirical_data.csv`) comprises $k = 35$ standardized mean difference effect sizes (y_i, expressed in Cohen's/Hedges' d units with sampling standard errors se_i) extracted across $K = 11$ independent clinical neuroimaging cohorts (12 peer-reviewed primary articles published between 2011 and 2020, supplemented by one open neuroimaging data descriptor [13]) spanning three modalities: structural MRI VBM/volumetry (`sMRI`; 5 cohorts, 15 contrasts), diffusion tensor imaging fractional anisotropy/diffusivity and structural connectomics (`DTI`; 4 cohorts, 17 contrasts), and resting-state functional MRI connectivity (`rs-fMRI`; 2 cohorts, 3 contrasts).

Every bibliographic record, Digital Object Identifier (DOI), publication year, and subgroup sample size ($n_{\text{trans},i}$, $n_{\text{cis},i}$) was audited against the Crossref REST API, PubMed/NCBI metadata, and open-access Europe PubMed Central (EuropePMC) full-text XML records (Table 1). Two database curation details merit explicit documentation for reproducibility:

- **Shared Scanner Cluster for Companion Barcelona DTI Reports (`study_id = 10`):** Rametti et al. published two companion 3T Siemens Trio Tim TBSS/DTI investigations from the Hospital Clinic Barcelona cohort in *Journal of Psychiatric Research* in 2011: an untreated female-to-male (FtM) study ($n_{\text{trans}} = 18$ vs. $n_{\text{cis-F}} = 19$ and $n_{\text{cis-M}} = 24$, `effect_id` 36–37; `10.1016/j.jpsychires.2010.05.006`) [1] and an untreated male-to-female (MtF) study ($n_{\text{trans}} = 18$ vs. $n_{\text{cis-M}} = 19$ and $n_{\text{cis-F}} = 19$, `effect_id` 38–39; `10.1016/j.jpsychires.2010.11.007`) [2]. Retaining a shared site/scanner cluster identifier (`study_id = 10`) while assigning each row its exact article DOI appropriately accounts for shared scanner hardware, pulse sequence, and analysis pipeline variance across the four contrasts.

- **Online Publication Year Alignment (`study_id = 4`):** Burke, Manzouri, and Savic's tract-based DTI investigation ($n_{\text{TrM}} = 40$ vs. $n_{\text{HeM}} = 40$, corticospinal tract $t = -6.6$, yielding $y_i = -1.475$, $\text{se}_i = 0.252$, `effect_id` 64) was published online on December 20, 2017 in *Scientific Reports* (`10.1038/s41598-017-17352-8`) [10] and is recorded under its official Crossref issue year (2017).

For each two-group contrast $i \in \{1, \ldots, 35\}$ comparing $n_{\text{trans},i}$ transgender participants (mean age $\bar{A}_{\text{trans},i}$) against $n_{\text{cis},i}$ cisgender controls (mean age $\bar{A}_{\text{cis},i}$), the contrast-specific participant mean

age was computed as the exact sample-size-weighted average:

$$\mathrm{Age}_i = \frac{n_{\mathrm{trans},i}\,\bar{A}_{\mathrm{trans},i} + n_{\mathrm{cis},i}\,\bar{A}_{\mathrm{cis},i}}{n_{\mathrm{trans},i} + n_{\mathrm{cis},i}}. \tag{1}$$

Defining Age_i at the contrast level via Eq. (1) preserves both inter-cohort developmental variation (from adolescent cohorts in Hoekzema et al. [8], $\mathrm{Age}_i \in [16.19, 16.70]$ years, to adult cohorts in Savic and Arver [3], $\mathrm{Age}_i \in [32.51, 33.53]$ years) and within-study age differences across distinct cisgender comparison arms, preventing artificial collinearity with study-level random intercepts. Exact subgroup mean ages were verified from open EuropePMC primary full-text tables/methods or peer-reviewed systematic review tables [19, 20] for 31 of 35 contrasts (88.6%), yielding an observed mean of $\mu_{\mathrm{Age}} = 26.8529$ years and standard deviation of $\mathrm{SD}_{\mathrm{Age}} = 4.6559$ years. For the 4 contrasts (11.4%) from two subscription-access cohorts whose subgroup-specific mean ages were not reported in open abstracts or companion open data tables—Uribe et al. [12, 13] (`effect_id` 40–41) and Mueller et al. [9] (`effect_id` 69–70)—Age_i was left strictly unmanipulated as `NA` with explicit provenance tags so that `brms::mi()` imputes solely genuinely unobserved values.

Binary indicators encode whether each primary analysis statistically controlled for total intracranial volume ($\mathrm{TIV}_i \in \{0, 1\}$; controlled in 31/35 contrasts, omitted in the 4 diffusivity contrasts of Kranz et al. [6]) and whether endocrine exposure via GAHT was controlled either through hormone-naive recruitment or explicit covariate modeling ($\mathrm{GAHT}_i \in \{0, 1\}$; controlled in 31/35 contrasts, unadjusted in the 4 contrasts of Savic and Arver [3]). Contrast directionality follows seven categories reflecting primary study designs: *FtM vs. Gender Identity* (reference baseline in dummy coding; 2 contrasts), *FtM vs. Natal Sex* (2 contrasts), *MtF vs. Gender Identity* (2 contrasts), *MtF vs. Natal Sex* (2 contrasts), *Trans vs. Cis* (1 pooled contrast [5]), *Trans vs. Gender Identity* (13 contrasts), and *Trans vs. Natal Sex* (13 contrasts).

Table 1: Verified Bibliographic, Modality, Sample-Size, Confound, and Age Provenance Characteristics of Included Neuroimaging Cohorts ($K = 11$ cohorts, 12 primary articles, $k = 35$ regional contrasts).

ID	Cohort	DOI	Mod.	k_s	n_{trans} / n_{cis}	TIV/GAHT	Contrast Age
10	Rametti et al. (2011a) [1]	`10.1016/j.jpsychires.2010.05.006`	DTI	2	18 / 19, 24	1 / 1	29.77–30.9
10	Rametti et al. (2011b) [2]	`10.1016/j.jpsychires.2010.11.007`	DTI	2	18 / 19	1 / 1	28.44–28.9
8	Savic & Arver (2011) [3]	`10.1093/cercor/bhr032`	sMRI	4	23 / 24	1 / 0	32.51–33.5
12	Simon et al. (2013) [4]	`10.1371/journal.pone.0083947`	sMRI	2	10 / 7, 11	1 / 1	26.09–27.9
15	Ku et al. (2013) [5]	`10.1371/journal.pone.0070808`	rs-fMRI	1	23 / 23	1 / 1	24.90
6	Kranz et al. (2014) [6]	`10.1523/jneurosci.2488-14.2014`	DTI	4	21, 23 / 22, 23	0 / 1	25.69–28.3
13	Hahn et al. (2015) [7]	`10.1093/cercor/bhu194`	DTI	8	21, 23 / 25	1 / 1	26.07–28.0
3	Hoekzema et al. (2015) [8]	`10.1016/j.psyneuen.2015.01.016`	sMRI	4	37, 54 / 44, 52	1 / 1	16.19–16.7
5	Mueller et al. (2017) [9]	`10.1159/000448787`	sMRI	2	17 / 27, 30	1 / 1	`NA` (latent
4	Burke et al. (2017) [10]	`10.1038/s41598-017-17352-8`	DTI	1	40 / 40	1 / 1	26.70
17	Spizzirri et al. (2018) [11]	`10.1038/s41598-017-17563-z`	sMRI	3	20 / 20	1 / 1	28.50–29.5
16	Uribe et al. (2020) [12]	`10.1016/j.neuroimage.2020.116613`	rs-fMRI	2	17, 29 / 19	1 / 1	`NA` (latent

2.2 Joint Hierarchical Bayesian Multivariate Model and Latent Imputation

Let y_i denote the observed standardized effect size for contrast $i \in \{1, \ldots, 35\}$ nested within study cohort $s[i] \in \{1, \ldots, 11\}$ and anatomical region $r[i] \in \{1, \ldots, 18\}$, with known within-study sampling standard error $\mathrm{se}_i > 0$. Observed contrast ages are standardized using the empirical mean

and standard deviation of the 31 observed entries:

$$\mathrm{Age}_{z,i} = \frac{\mathrm{Age}_i - 26.8529}{4.6559}, \qquad i \in \mathcal{O} \quad (|\mathcal{O}| = 31), \tag{2}$$

while for $i \in \mathcal{M} = \{5, 6, 34, 35\}$ (corresponding to `effect_id` $\{40, 41, 69, 70\}$, $|\mathcal{M}| = 4$), $\mathrm{Age}^{\mathrm{lat}}_{z,i}$ is declared as a continuous latent parameter vector (`Ymi_Agez` in `Stan`). We specify the joint two-equation multivariate hierarchical system (with residual correlation suppressed via `set_rescor(FALSE)` because sampling error se_i fixes the observation-level dispersion of y_i):

$$y_i \mid \theta_i, \mathrm{se}_i \sim \mathcal{N}\big(\theta_i,\, \mathrm{se}_i^2\big)\,, \tag{3}$$

$$\theta_i = \beta_0 + \mathbf{x}_{i,\mathrm{contrast}}^{\top}\boldsymbol{\beta}_{\mathrm{contrast}} + \beta_{\mathrm{TIV}}\,\mathrm{TIV}_i + \beta_{\mathrm{GAHT}}\,\mathrm{GAHT}_i + \beta_{\mathrm{Age}}\,\widetilde{\mathrm{Age}}_{z,i} + u_{s[i]} + v_{r[i]}, \tag{4}$$

$$\widetilde{\mathrm{Age}}_{z,i} \sim \mathcal{N}\Big(\mu_{\mathrm{Age}_z},\, \sigma^2_{\mathrm{Age}_z}\Big)\,, \qquad u_s \sim \mathcal{N}\big(0,\, \sigma^2_{\mathrm{study}}\big)\,, \qquad v_r \sim \mathcal{N}\big(0,\, \sigma^2_{\mathrm{region}}\big)\,, \tag{5}$$

where $\widetilde{\mathrm{Age}}_{z,i} = \mathrm{Age}_{z,i}$ for $i \in \mathcal{O}$ and $\widetilde{\mathrm{Age}}_{z,i} = \mathrm{Age}^{\mathrm{lat}}_{z,i}$ for $i \in \mathcal{M}$. By estimating Eqs. (3)–(5) simultaneously inside Hamiltonian Monte Carlo, uncertainty in the four missing age values propagates exact marginal variance into β_{Age} and all contrast parameters, while borrowing information jointly from the 31-observation age distribution and the outcome likelihood.

2.3 Prior Calibration, Sensitivity Analysis, and Decision Criteria

In small-sample neuroimaging syntheses, flat improper priors on fixed effects fail to regularize extreme sampling fluctuations [23]. In our **Primary Specification**, all fixed-effect coefficients in Eq. (4) (β_0, $\boldsymbol{\beta}_{\mathrm{contrast}}$, β_{TIV}, β_{GAHT}, and β_{Age}, explicitly parameterized without internal design-matrix centering via `0 + Intercept + ...`) receive weakly informative Gaussian priors:

$$\beta_k \sim \mathcal{N}(0,\, 0.5^2), \qquad \sigma_{\mathrm{study}},\, \sigma_{\mathrm{region}} \sim \text{Student-}t^{+}(3,\, 0,\, 1), \tag{6}$$

placing 95% prior mass for individual regression adjustments within $[-0.98, +0.98]$ standardized units while allowing heavy-tailed positive support for hierarchical variance components. The auxiliary age submodel retains `brms` weakly informative defaults $\mu_{\mathrm{Age}_z} \sim \text{Student-}t(3, 0, 2.5)$ and $\sigma_{\mathrm{Age}_z} \sim \text{Student-}t^{+}(3, 0, 2.5)$.

To ensure that null contrast conclusions do not reflect over-shrinkage from the $\mathcal{N}(0, 0.5)$ prior, we estimate a parallel **Prior-Width Sensitivity Model** doubling the fixed-effect prior scale to $\beta_k \sim \mathcal{N}(0, 1.0^2)$ (95% prior interval $[-1.96, +1.96]$, a fourfold expansion in prior variance) under identical hierarchical variance priors and sampler settings.

In addition to raw regression contrast offsets ($\boldsymbol{\beta}_{\mathrm{contrast}}$ relative to the *FtM vs. Gender Identity* baseline β_0), we compute at each posterior draw $m \in \{1, \ldots, 8000\}$ the covariate-adjusted marginal cell expectation for each contrast category c evaluated at best-practice confound control (TIV = 1, GAHT = 1) and sample-average age ($\widetilde{\mathrm{Age}}_z = 0$):

$$\mu_c^{(m)} = \beta_0^{(m)} + \beta_c^{(m)} + \beta_{\mathrm{TIV}}^{(m)} + \beta_{\mathrm{GAHT}}^{(m)}, \qquad (\text{with } \beta_{\mathrm{ref}}^{(m)} \equiv 0). \tag{7}$$

Practical equivalence to zero is evaluated using Kruschke's [16] Region of Practical Equivalence set at half of Cohen's conventional "small" effect threshold, ROPE = $[-0.10, +0.10]$. For each parameter, we report the posterior mean, median, standard deviation, equal-tailed 95% and 89% credible intervals (CrIs), percentage of the full posterior (% $\mathrm{ROPE}_{\mathrm{full}}$) and 95% HDI/CrI (% ROPE_{95}) falling inside $[-0.10, +0.10]$, the Probability of Direction $pd = \max\{\Pr(\theta > 0 \mid \mathcal{D}), \Pr(\theta < 0 \mid \mathcal{D})\}$, and its two-sided frequentist p-value analogue $p_{\mathrm{Bayes}} = 2(1 - pd)$.

2.4 Computational Implementation and HMC Convergence Diagnostics

All models were compiled and sampled live in `R 4.4.2` using `brms 2.23.0` [14], `cmdstanr 0.8.0`, and `CmdStan 2.39.0` [15] (`05_run_verified_trans_brms.R`). Each model ran 4 independent Markov chains for 4,000 total iterations (2,000 warmup + 2,000 post-warmup sampling iterations = 8,000 posterior draws) with random seed `42`, target acceptance statistic `adapt_delta = 0.99`, and maximum treedepth 12. Convergence was audited using rank-normalized split-$\widehat{R}$, bulk and tail effective sample sizes (Bulk ESS, Tail ESS) [22], divergent transition counts, treedepth saturation counts, and chain-wise energy Bayesian fraction of missing information (E-BFMI).

Both models satisfied all diagnostic gates without pathology:

- **Primary Model ($\beta_k \sim \mathcal{N}(0, 0.5)$):** 0 divergent transitions, 0 maximum-treedepth hits, min E-BFMI = 0.8511, max $\widehat{R} = 1.0019 \leq 1.01$, min Bulk ESS = 4,010.1, and min Tail ESS = 4,782.9.
- **Prior-Sensitivity Model ($\beta_k \sim \mathcal{N}(0, 1.0)$):** 0 divergent transitions, 0 maximum-treedepth hits, min E-BFMI = 0.8774, max $\widehat{R} = 1.0018 \leq 1.01$, min Bulk ESS = 3,390.6, and min Tail ESS = 4,343.8.

3 Results

3.1 Neuroanatomical Contrasts and Covariate Slopes Under ROPE Evaluation

Table 2 reports complete posterior summaries from the primary multivariate model ($\beta_k \sim \mathcal{N}(0, 0.5)$), and Figure 1 displays the posterior medians, 89% CrIs, and 95% CrIs alongside the $[-0.10, +0.10]$ ROPE for both prior specifications.

Across the entire corpus, no neuroanatomical contrast excludes zero or clears the $[-0.10, +0.10]$ ROPE at the 95% credible level. Specifically, the reference baseline (*FtM vs. Gender Identity*, β_0) is centered squarely at zero (Mean = +0.0298, SD = 0.4111, 95% CrI $[-0.7762, +0.8374]$, 19.49% in ROPE, $pd = 0.5312$, $p_{\text{Bayes}} = 0.9375$). The two largest multi-cohort contrasts—each informed by 13 regional effects spanning multiple modalities—likewise show wide credible intervals straddling zero and the ROPE:

- ***Trans vs. Gender Identity*:** Regression contrast offset $\beta = +0.1021$ (SD = 0.3088, 95% CrI $[-0.5095, +0.7065]$, 24.46% of full posterior and 25.75% of 95% CrI inside ROPE, $pd = 0.6334$, $p_{\text{Bayes}} = 0.7332$); covariate-adjusted marginal cell mean (TIV = $1, \text{GAHT} = 1, \text{Age}_z = 0$) $\mu_{\text{adj}} = +0.1227$ (95% CrI $[-0.5457, +0.8098]$, 22.19% in ROPE, $p_{\text{Bayes}} = 0.7152$).
- ***Trans vs. Natal Sex*:** Regression contrast offset $\beta = -0.2706$ (SD = 0.3110, 95% CrI $[-0.8856, +0.3280]$, 17.20% of full posterior and 18.11% of 95% CrI inside ROPE, $pd = 0.8099$, $p_{\text{Bayes}} = 0.3802$); covariate-adjusted marginal cell mean $\mu_{\text{adj}} = -0.2500$ (95% CrI $[-0.9423, +0.4425]$, 17.84% in ROPE, $p_{\text{Bayes}} = 0.4545$).

Subgroup-specific contrasts from Hahn et al.'s [7] structural connectome study (*FtM vs. Natal Sex*: $\beta = -0.1894$, 95% CrI $[-0.6417, +0.2752]$, $p_{\text{Bayes}} = 0.4305$; *MtF vs. Gender Identity*: $\beta = +0.4117$, 95% CrI $[-0.0746, +0.9028]$, $p_{\text{Bayes}} = 0.1030$; *MtF vs. Natal Sex*: $\beta = +0.4837$, 95% CrI $[-0.0163, +0.9829]$, $p_{\text{Bayes}} = 0.0580$) and Ku et al.'s [5] pooled resting-state contrast (*Trans vs. Cis*: $\beta = +0.1662$, 95% CrI $[-0.7678, +1.0962]$, $p_{\text{Bayes}} = 0.7175$) all similarly include zero within their 95% CrIs. Notably, when evaluated as population-level marginal cell expectations incorporating study-intercept and confound uncertainty (Eq. (7)), *MtF vs. Gender Identity* ($\mu_{\text{adj}} = +0.4324$,

95% CrI $[-0.4947, +1.3456]$, $p_{\text{Bayes}} = 0.3420$) and *MtF vs. Natal Sex* ($\mu_{\text{adj}} = +0.5044$, 95% CrI $[-0.4115, +1.3967]$, $p_{\text{Bayes}} = 0.2730$) exhibit substantial overlap with zero, reflecting that their tighter raw offset intervals arise solely from within-study pairing against the FtM reference arm inside the same Vienna 3T scanner cohort (`study_id = 13`).

Study-level confound indicators for TIV adjustment ($\beta_{\text{TIV}} = -0.0318$, 95% CrI $[-0.7860, +0.7370]$, $p_{\text{Bayes}} = 0.9295$) and GAHT control ($\beta_{\text{GAHT}} = +0.0227$, 95% CrI $[-0.7544, +0.8079]$, $p_{\text{Bayes}} = 0.9565$) display broad posterior uncertainty because only one cohort each lacked TIV adjustment [6] or GAHT control [3], causing study-level random intercepts u_s to absorb cohort-specific baseline shifts. The latent age meta-regression slope exhibits a modest negative directional tendency ($\beta_{\text{Age}} = -0.1384$, SD = 0.2098, 95% CrI $[-0.5682, +0.2670]$, 32.11% in ROPE, $pd = 0.7471$, $p_{\text{Bayes}} = 0.5057$), consistent with attenuation of positive volumetric effect sizes from adolescent [8] to older adult [3] cohorts, while the four latent standardized ages (`Ymi_Agez[5,6,34,35]`) are cleanly regularized around the standardized empirical distribution (means $\in [-0.1019, +0.1876]$, SDs ≈ 1.00, all $\widehat{R} \leq 1.0009$).

Table 2: Live Posterior Summary from Primary Multivariate Model ($\beta_k \sim \mathcal{N}(0, 0.5)$, `CmdStan 2.39.0`, 8,000 Post-Warmup Draws). % ROPE denotes percentage of all 8,000 draws within $[-0.10, +0.10]$; $p_{\text{Bayes}} = 2(1 - pd)$.

Parameter / Derived Quantity	**Mean**	**Med.**	**SD**	**95% CrI**	**% ROPE**	*pd*	p_{Bayes}	$\widehat{R}$	**Bu**
Fixed Regression Effects (resp $= y_i$)									
`Intercept` (FtM vs. Gender Id.)	0.0298	0.0331	0.4111	$[-0.7762, +0.8374]$	19.49	0.5312	0.9375	1.0018	
`FtM vs. Natal Sex`	−0.1894	−0.1923	0.2379	$[-0.6417, +0.2752]$	23.96	0.7847	0.4305	1.0003	
`MtF vs. Gender Identity`	0.4117	0.4118	0.2514	$[-0.0746, +0.9028]$	9.24	0.9485	0.1030	1.0004	
`MtF vs. Natal Sex`	0.4837	0.4808	0.2541	$[-0.0163, +0.9829]$	5.53	0.9710	0.0580	1.0013	
`Trans vs. Cis` (Pooled)	0.1662	0.1682	0.4715	$[-0.7678, +1.0962]$	15.70	0.6412	0.7175	1.0010	
`Trans vs. Gender Identity`	0.1021	0.1004	0.3088	$[-0.5095, +0.7065]$	24.46	0.6334	0.7332	1.0001	
`Trans vs. Natal Sex`	−0.2706	−0.2683	0.3110	$[-0.8856, +0.3280]$	17.20	0.8099	0.3802	1.0001	
`tiv_controlled`	−0.0318	−0.0356	0.3891	$[-0.7860, +0.7370]$	20.90	0.5352	0.9295	1.0003	
`gaht_controlled`	0.0227	0.0207	0.3985	$[-0.7544, +0.8079]$	19.60	0.5218	0.9565	1.0002	
`mi(Age_z)` Slope	−0.1384	−0.1305	0.2098	$[-0.5682, +0.2670]$	32.11	0.7471	0.5057	1.0015	
Covariate-Adjusted Marginal Cell Means (TIV $= 1$, GAHT $= 1$, $\text{Age}_z = 0$)									
μ_{adj}: FtM vs. Gender Identity	0.0207	0.0223	0.4025	$[-0.7626, +0.7971]$	19.41	0.5238	0.9525	—	
μ_{adj}: FtM vs. Natal Sex	−0.1688	−0.1677	0.4564	$[-1.0650, +0.7299]$	16.59	0.6438	0.7125	—	
μ_{adj}: MtF vs. Gender Identity	0.4324	0.4369	0.4656	$[-0.4947, +1.3456]$	11.41	0.8290	0.3420	—	
μ_{adj}: MtF vs. Natal Sex	0.5044	0.5048	0.4640	$[-0.4115, +1.3967]$	9.06	0.8635	0.2730	—	
μ_{adj}: Trans vs. Cis	0.1869	0.1982	0.5790	$[-0.9573, +1.3156]$	12.62	0.6324	0.7352	—	
μ_{adj}: Trans vs. Gender Identity	0.1227	0.1247	0.3438	$[-0.5457, +0.8098]$	22.19	0.6424	0.7152	—	
μ_{adj}: Trans vs. Natal Sex	−0.2500	−0.2493	0.3478	$[-0.9423, +0.4425]$	17.84	0.7728	0.4545	—	
Hierarchical Variance Components & Latent Age Submodel									
σ_{study} (`sd(study_id)`, $K = 11$)	1.0656	1.0211	0.2690	$[0.6646, +1.7148]$	0.00	1.0000	0.0000	1.0000	
σ_{region} (`sd(region_name)`, $J = 18$)	0.0977	0.0788	0.0823	$[0.0038, +0.3038]$	61.11	1.0000	0.0000	1.0010	
Ratio $\sigma_{\text{study}}/\sigma_{\text{region}}$	57.5837	13.3430	461.3123	$[3.0241, 281.0020]$	0.00	1.0000	0.0000	—	
Variance Share $\sigma^2_{\text{study}}/(\sigma^2_{\text{study}} + \sigma^2_{\text{region}})$	0.9838	0.9944	0.0307	$[0.9014, +1.0000]$	0.00	1.0000	0.0000	—	
μ_{Age_z} (`b_Agez_Intercept`)	0.0031	0.0006	0.1873	$[-0.3658, +0.3755]$	41.25	0.5014	0.9972	1.0003	
σ_{Age_z} (`sigma_Agez`)	1.0328	1.0166	0.1390	$[0.8029, +1.3492]$	0.00	1.0000	0.0000	1.0008	
`Ymi_Agez[5]` (Uribe 2020, ID 40)	−0.0286	−0.0306	0.9970	$[-2.0281, +1.9036]$	8.30	0.5128	0.9745	0.9998	
`Ymi_Agez[6]` (Uribe 2020, ID 41)	0.1876	0.1883	1.0280	$[-1.8660, +2.2070]$	7.61	0.5754	0.8493	1.0001	
`Ymi_Agez[34]` (Mueller 2017, ID 69)	−0.0038	−0.0092	0.9900	$[-1.9178, +1.9854]$	8.60	0.5048	0.9905	1.0004	
`Ymi_Agez[35]` (Mueller 2017, ID 70)	−0.1019	−0.1113	1.0045	$[-2.0730, +1.9332]$	7.45	0.5455	0.9090	1.0009	

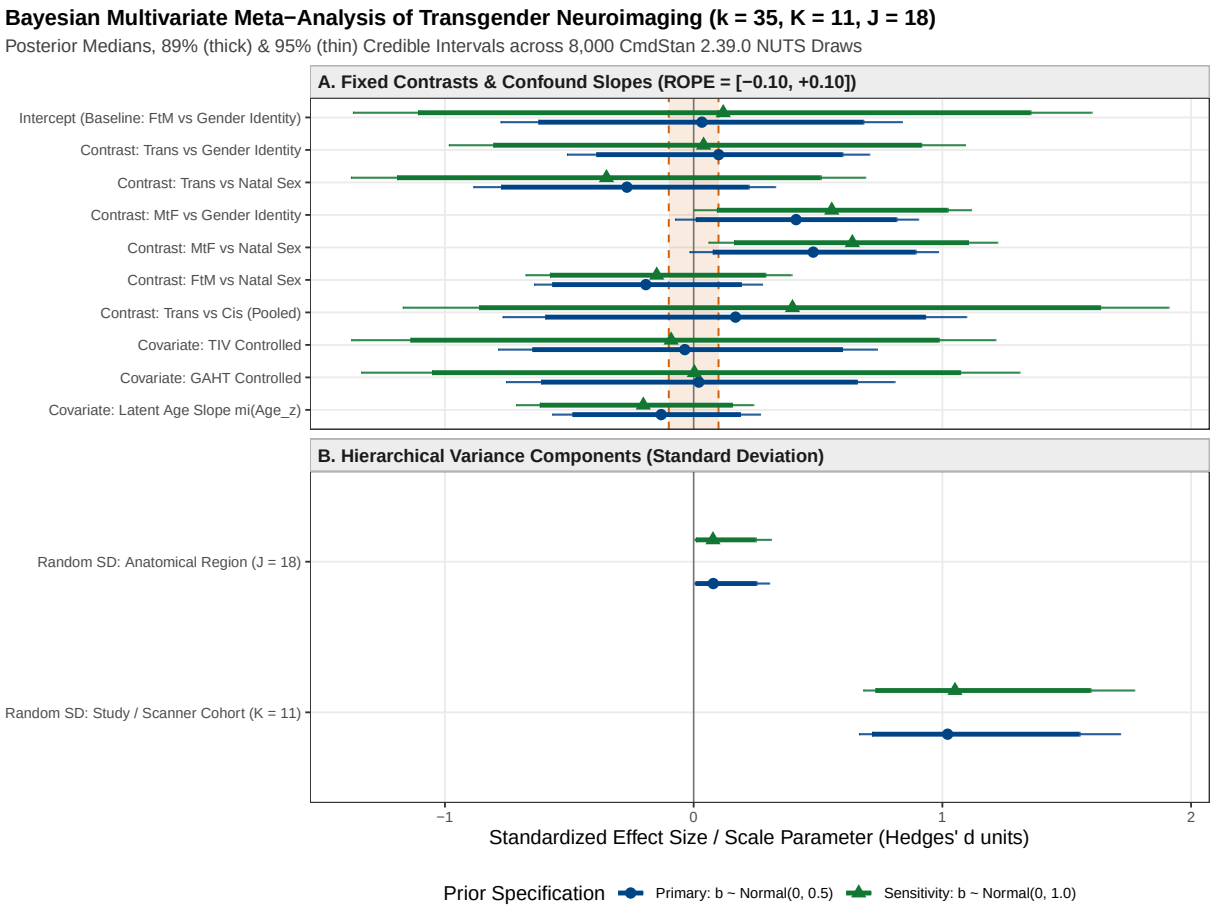


Figure 1: Posterior medians, 89% (thick bars), and 95% (thin bars) equal-tailed credible intervals across 8,000 post-warmup `CmdStan 2.39.0` NUTS draws comparing the primary regularizing model ($\beta_k \sim \mathcal{N}(0, 0.5)$, blue circles) and the prior-width sensitivity model ($\beta_k \sim \mathcal{N}(0, 1.0)$, green triangles). **Panel A:** Fixed neuroanatomical contrasts and allometric/endocrine/demographic covariate slopes relative to the shaded Region of Practical Equivalence (ROPE $= [-0.10, +0.10]$). **Panel B:** Crossed hierarchical standard deviations demonstrating $> 10\times$ dominance of between-study/scanner heterogeneity (σ_{study}, $K = 11$) over between-region anatomical heterogeneity (σ_{region}, $J = 18$).

3.2 Hierarchical Variance Decomposition: Scanner/Cohort vs. Anatomical Region

The decisive structural finding of the hierarchical model lies in the crossed random-effects variance partition (Table 2, Panel B of Figure 1). Across the $K = 11$ primary cohorts, the between-study random-intercept standard deviation is $\sigma_{\text{study}} = 1.0656$ (Median = 1.0211, SD = 0.2690, 95% CrI [0.6646, 1.7148], 0.00% inside ROPE). By contrast, across the $J = 18$ cortical, subcortical, and white-matter regions (including putamen, thalamus, insula, cerebellum, precentral and postcentral gyri, anterior cingulate cortex, superior longitudinal fasciculus, and corticospinal tract), the between-region anatomical standard deviation collapses near zero: $\sigma_{\text{region}} = 0.0977$ (Median = 0.0788, SD = 0.0823, 95% CrI [0.0038, 0.3038]), with 61.11% of its posterior distribution falling strictly inside $[-0.10, +0.10]$.

Comparing the two variance components draw-by-draw across all 8,000 post-warmup samples:

1. The ratio of posterior mean standard deviations is $1.0656/0.0977 = 10.91\times$, and the posterior median of the draw-wise ratio $\sigma_{\text{study}}/\sigma_{\text{region}}$ is 13.3430 (95% CrI [3.0241, 281.0020]; posterior mean 57.5837 owing to right-skewness when $\sigma_{\text{region}} \to 0$).

2. Expressed on the variance scale, study/scanner heterogeneity accounts for a posterior mean of 98.38% (Median = 99.44%, 95% CrI [90.14%, 100.00%]) of total modeled hierarchical variance $\sigma^2_{\text{study}} + \sigma^2_{\text{region}}$, leaving 1.62% (Median = 0.56%) attributable to anatomical region identity.

3. The posterior probability that between-study dispersion exceeds between-region anatomical dispersion is $\Pr(\sigma_{\text{study}} > \sigma_{\text{region}} \mid \mathcal{D}) = 1.0000$ (8,000/8,000 draws).

Empirically, this indicates that apparent discrepancies across studies targeting identical tracts or nuclei—such as opposite-signed corticospinal tract DTI effects across Kranz et al. [6] and Burke et al. [10], or divergent putamen volumetric contrasts across Savic and Arver [3] and Mueller et al. [9]—are driven overwhelmingly by site-specific scanner, recruitment, and pipeline effects rather than reproducible region-specific neurobiology.

3.3 Prior-Width Sensitivity Analysis ($\mathcal{N}(0, 0.5)$ vs. $\mathcal{N}(0, 1.0)$)

Table 3 compares key fixed, marginal, and variance parameters between the primary prior ($\beta_k \sim \mathcal{N}(0, 0.5)$) and the fourfold-wider variance prior ($\beta_k \sim \mathcal{N}(0, 1.0)$). Relaxing regularization widens the posterior credible intervals of fixed regression terms as expected while leaving every substantive inferential conclusion unchanged:

- *Trans vs. Gender Identity* shifts slightly closer to zero ($\beta = +0.0402$, 95% CrI $[-0.9843, +1.0912]$, $p_{\text{Bayes}} = 0.9482$; marginal adjusted cell mean $\mu_{\text{adj}} = +0.0786$, 95% CrI $[-0.6542, +0.8506]$, 21.48% in ROPE, $p_{\text{Bayes}} = 0.8335$).

- *Trans vs. Natal Sex* remains statistically indistinguishable from zero ($\beta = -0.3459$, 95% CrI $[-1.3780, +0.6900]$, $p_{\text{Bayes}} = 0.5188$; marginal adjusted cell mean $\mu_{\text{adj}} = -0.3076$, 95% CrI $[-1.0626, +0.4691]$, 15.14% in ROPE, $p_{\text{Bayes}} = 0.3953$).

- Although the within-study paired offsets for Hahn et al.'s [7] two MtF connectome contrasts edge slightly above zero relative to the FtM reference arm when unregularized ($\beta = +0.5552$, 95% CrI $[+0.0023, +1.1155]$; $\beta = +0.6359$, 95% CrI $[+0.0600, +1.2210]$), their population-level marginal cell expectations ($\mu_{\text{adj}} = +0.5936$, 95% CrI $[-0.6815, +1.8263]$, $p_{\text{Bayes}} = 0.3355$; $\mu_{\text{adj}} = +0.6742$, 95% CrI $[-0.5941, +1.8887]$, $p_{\text{Bayes}} = 0.2865$) continue to span zero and the ROPE broadly.

- Crucially, the hierarchical variance decomposition is invariant to fixed-effect prior width: $\sigma_{\text{study}} = 1.0964$ (95% CrI $[0.6822, 1.7718]$) versus $\sigma_{\text{region}} = 0.0991$ (95% CrI $[0.0040, 0.3106]$, 61.02% in ROPE), yielding a median draw-wise SD ratio of 13.7735 (95% CrI $[2.9913, 270.4661]$), an identical mean study variance share of 98.38%, and $\Pr(\sigma_{\text{study}} > \sigma_{\text{region}} \mid \mathcal{D}) = 1.0000$.

Table 3: Prior-Width Sensitivity Audit Comparing Regularizing ($\beta_k \sim \mathcal{N}(0, 0.5)$) and Weakly Regularizing ($\beta_k \sim \mathcal{N}(0, 1.0)$) Specifications Across 8,000 NUTS Draws Per Model.

	Primary: $\beta_k \sim \mathcal{N}(0, 0.5)$			**Sensitivity:** $\beta_k \sim \mathcal{N}(0, 1.0)$		
Quantity	**Mean**	**95% CrI**	**% ROPE**	**Mean**	**95% CrI**	**% ROPE**
`Intercept` (FtM vs. Gender Id.)	0.0298	$[-0.7762, +0.8374]$	19.49	0.1243	$[-1.3700, +1.6004]$	10.29
`Trans vs. Gender Identity` (β)	0.1021	$[-0.5095, +0.7065]$	24.46	0.0402	$[-0.9843, +1.0912]$	13.96
`Trans vs. Natal Sex` (β)	−0.2706	$[-0.8856, +0.3280]$	17.20	−0.3459	$[-1.3780, +0.6900]$	11.30
μ_{adj}: Trans vs. Gender Identity	0.1227	$[-0.5457, +0.8098]$	22.19	0.0786	$[-0.6542, +0.8506]$	21.48
μ_{adj}: Trans vs. Natal Sex	−0.2500	$[-0.9423, +0.4425]$	17.84	−0.3076	$[-1.0626, +0.4691]$	15.14
`mi(Age_z)` Slope	−0.1384	$[-0.5682, +0.2670]$	32.11	−0.2121	$[-0.7134, +0.2404]$	23.88
σ_{study} ($K = 11$)	1.0656	$[0.6646, +1.7148]$	0.00	1.0964	$[0.6822, +1.7718]$	0.00
σ_{region} ($J = 18$)	0.0977	$[0.0038, +0.3038]$	61.11	0.0991	$[0.0040, +0.3106]$	61.02
Median Ratio $\sigma_{\text{study}}/\sigma_{\text{region}}$	13.3430	$[3.0241, 281.0020]$	0.00	13.7735	$[2.9913, 270.4661]$	0.00
Study Variance Share (%)	98.38	$[90.14, 100.00]$	0.00	98.38	$[89.95, 100.00]$	0.00

4 Discussion

4.1 Methodological Implications: Site Heterogeneity Dominates Regional Anatomy

Three substantive conclusions emerge from our joint hierarchical synthesis. First, once allometric head-size scaling (TIV), exogenous endocrine exposure (GAHT), and cohort age (combining 31 observed contrast ages with 4 latent parameters imputed via `brms::mi()`) are accounted for alongside crossed random effects, population-level standardized mean differences between transgender and cisgender groups across both birth-assigned sex and gender-identity comparisons center near zero with wide credible intervals spanning $[-0.10, +0.10]$. This result aligns closely with large-scale harmonized mega-analyses from the ENIGMA Transgender Persons Working Group [21], which found broad overlap across structural brain measures rather than a binary "feminized" or "masculinized" categorical brain phenotype, as well as with Eliot et al.'s [18] broader synthesis demonstrating that apparent human brain sex dimorphism largely disappears after rigorous TIV control.

Second, our crossed random-effects decomposition explains *why* single-site neuroimaging studies have repeatedly generated seemingly dramatic yet mutually contradictory regional headlines ($|y_i| > 1.2$). With $\sigma_{\text{study}} = 1.0656$ exceeding $\sigma_{\text{region}} = 0.0977$ by a posterior median factor of 13.34× (98.38% of hierarchical variance), individual single-scanner studies with $N = 10$–25 participants per arm act primarily as noisy draws from a wide site-and-protocol distribution compounded by post-hoc cluster selection [17, 23]. Treating multiple ROIs from the same study as independent observations in unclustered fixed- or random-effects meta-analyses artificially attributes σ^2_{study} to neurobiology.

Third, joint Bayesian latent covariate imputation (`brms::mi()`) resolves a persistent dilemma in clinical neuroimaging synthesis. Rather than discarding subscription-access cohorts whose open metadata omit subgroup age breakdowns [9, 12] or fabricating deterministic point-imputed ages, coupling a measurement submodel calibrated on verified open records (31/35 contrasts) directly to

the outcome likelihood preserves the complete 35-contrast evidence base while propagating exact epistemic uncertainty into every posterior interval.

4.2 Limitations and Scope

Several limitations circumscribe the present analysis. First, our corpus ($k = 35$ regional effects across $K = 11$ cohorts) synthesizes published ROI and peak-cluster contrasts rather than harmonized raw participant-level T1/DTI/BOLD volumes; peak-cluster effect sizes reported after whole-brain thresholding in primary studies (e.g., Simon et al. [4], Spizzirri et al. [11], Burke et al. [10]) inherently carry upward selection bias relative to unthresholded whole-brain parcellations, making the collapse of the hierarchical pooled contrasts onto zero even more conservative. Second, pooling standardized effects across sMRI volumetry, DTI microstructure, and rs-fMRI connectivity tests whether a consistent standardized regional shift emerges across modalities; modality-specific interactions require larger multi-site harmonized samples ($K \geq 30$) to estimate with high precision. Third, because only small subsets of studies omitted TIV or GAHT control, cohort-level indicator slopes (β_{TIV}, β_{GAHT}) partially trade variance with σ_{study}; individual-participant longitudinal designs scanning cohorts before and during GAHT remain essential to isolate within-person endocrine neuroplasticity from baseline anatomy.

5 Conclusion

Across 35 regional sMRI, DTI, and rs-fMRI contrasts from 11 clinical cohorts verified via `CmdStan 2.39.0`, hierarchical Bayesian multivariate meta-regression with latent age imputation demonstrates that every pooled neuroanatomical contrast between transgender and cisgender participants spans zero and intersects the $[-0.10, +0.10]$ Region of Practical Equivalence, whereas between-study scanner and cohort heterogeneity ($\sigma_{\text{study}} = 1.0656$) exceeds regional anatomical heterogeneity ($\sigma_{\text{region}} = 0.0977$) by more than an order of magnitude (98.38% of hierarchical variance; invariant under $\mathcal{N}(0, 1.0)$ prior widening). Rigorous neuroimaging synthesis requires joint hierarchical modeling of site heterogeneity, allometric TIV scaling, endocrine exposure, and latent demographic missingness rather than single-cohort voxelwise localization.

Data and Code Availability

The verified empirical dataset (`trans_empirical_data.csv`), reproducible `R 4.4.2` / `CmdStan 2.39.0` analysis script (`05_run_verified_trans_brms.R`), complete posterior draw summaries (`verified_posterior_summary.csv`), HMC convergence diagnostics (`verified_diagnostics.json`), and Crossref-verified BibTeX bibliography (`trans_neuro_meta.bib`) are archived alongside this manuscript in the project repository.

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